

ChemNovus — Product Spec Sheet (fictional)

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FlexiBond HT

Product family: FlexiBond structural adhesives **Grade:** HT (high-temperature, two-component epoxy) **Form:** 2K epoxy, 50 mL cartridges / 200 L drums (Part A + Part B) **Product code:** FB-HT-A / FB-HT-B · Mix ratio 2:1 by volume

Description

Two-component, heat-resistant structural epoxy adhesive. High shear strength and thermal endurance for bonding dissimilar substrates (metal, ceramic, FR4, thermoset composites). Formulated for electronics assembly and EV battery-pack bonding where thermal cycling and vibration are severe.

Key properties (typical, cured)

Property	Value	Method
Chemistry	2K epoxy	—
Color (A / B)	grey / black	—
Mix viscosity (25 °C)	45,000 mPa·s	ISO 3219
Pot life (25 °C)	45 min	—
Fixture time	3 h	—
Full cure	24 h @ 23 °C or 45 min @ 80 °C	—
Lap shear strength (Al, 23 °C)	28 MPa	ISO 4587
Lap shear @ 150 °C	18 MPa	ISO 4587

Property	Value	Method
Operating temp range	−40 °C to +150 °C	—
Glass transition (T _g)	160 °C	DSC
Thermal conductivity	0.8 W/m·K	—
Dielectric strength	18 kV/mm	IEC 60243
Volume resistivity	1×10 ¹⁴ Ω·cm	—
Flammability	UL 94 V-0	—

Processing

- Static-mix nozzle or metered 2K dispenser
- Surface prep: degrease + abrade; primer optional on low-energy substrates
- Heat-accelerated cure recommended for line throughput

Applications

EV battery cell-to-module & module-to-pack bonding, busbar/potting, power-electronics heat-sink bonding, PCB component staking, motor magnet bonding, sensor encapsulation.

Compliance

REACH compliant · RoHS compliant · UL 94 V-0 · Halogen-free · Low-outgassing (ASTM E595 pass) · SDS available (2-part). **Not** medical/implant grade.

Commercial Notes

Best-fit customers: EV battery-pack assemblers, power-electronics & inverter makers, EMS/electronics assembly, e-motor manufacturers. Volume sweet spot 0.5–50 t/yr; often pulls technical-service engagement (bonding process design).